

AI/ML-Enhanced L1/L2 Triggered Mobility (LTM)

Technical Deep-Dive: MAC/UPC Scheduler Impact & Handover Interruption Time (HIT) Gains
Based on 3GPP TR 38.745 (Release 20 Study)

Executive Summary

Integrating Artificial Intelligence (AI) and Machine Learning (ML) predictive execution models with Layer 1/Layer 2 Triggered Mobility (LTM) represents an evolutionary leap toward zero-latency handovers in next-generation radio access networks (NG-RAN). By moving the handover execution from the reactive Radio Resource Control (RRC) layer (Layer 3) to the predictive and lower physical/MAC layers (Layers 1/2), the network significantly mitigates packet drops and communication gaps.

1. Operational Impact on the MAC Scheduler

The Medium Access Control (MAC) Scheduler allocates physical layer time-frequency resources to active User Equipments (UEs) on both uplink (UL) and downlink (DL). Under 3GPP TR 38.745 framework guidelines, AI/ML models alter traditional MAC scheduling paradigms in three major ways:

Predictive Resource & Buffer Allocation: Instead of responding blindly to post-handover access triggers, the target gNB's MAC Scheduler utilizes multi-hop trajectory forecasting to pre-allocate dedicated scheduling blocks and prioritize the incoming UE before it formally connects.

Optimized MAC Control Element (MAC CE) Transmission: Predictive filters isolate fast-fading channel variations, assisting the source base station to optimally time the transmission of the critical LTM cell-switch MAC CE command before deep signal degradation occurs.

Dynamic Beam State Mapping: Real-time AI prediction identifies the exact target Transmission Configuration Indicator (TCI) state, allowing the target MAC scheduler to immediately execute DL/UL transmissions without dead-time required for traditional beam sweeps.

2. Operational Impact on the UPC Scheduler

The User Plane Control (UPC) Scheduler manages packet classification, buffering, and Quality of Service (QoS) guarantees over split-architectures (gNB-CU/DU). AI/ML integration provides crucial timing advantages:

Proactive Packet Bi-Casting: By forecasting the target cell transition ahead of time, the UPC scheduler initiates selective packet forwarding or dual-connectivity bi-casting to the target candidate cell in advance. This ensures user data resides at the target buffer exactly when the lower-layer cell switch fires.

Reduced PDCP/RLC Recovery Overheads: Because AI-driven LTM ensures highly synchronized, seamless switching windows, fewer data frames are dropped or misrouted. This significantly minimizes Packet Data Convergence Protocol (PDCP) data recovery and Radio Link Control (RLC) retransmission loops.

3. Handover Interruption Time (HIT) Comparison

By combining lower-layer LTM execution with AI/ML prediction engines, the Time-To-Trigger (TTT) window is optimized or bypassed entirely, shrinking the latency gap down to sub-10ms intervals.

Mobility Protocol	Handover Interruption Time (HIT)	Core Driving Mechanism
Legacy L3 Handover	50ms - 90ms	Reactive L3 RRC Signaling, measurement reports, full protocol stack reset, and post-switch Random Access.
Standard L1/L2 LTM	20ms - 30ms	Target cells are preconfigured. Switching relies on fast lower-layer MAC CE signaling with early DL/UL synchronization.
AI/ML-Enhanced LTM	~0ms - Sub-10ms	AI models completely bypass the standard Time-To-Trigger (TTT) window. Proactive target preparation ensures near-instantaneous MAC-CE execution.